

MODULE 7: FIREWALL

Contents

Firewall	1
Types of Firewalls	2
Firewall Configurations	3
DMZ (Demilitarized Zone) Network	4
DMZ Architecture	5

Firewall

A firewall is a hardware device, software application, or combination of both that enforces security policies by permitting or denying network traffic. It acts like a network security system that monitors, filters, and controls incoming and outgoing network traffic based on predefined security rules.

It acts as a **barrier between a trusted internal network and an untrusted external network (such as the Internet)**. The primary purpose of a firewall is to prevent unauthorized access while allowing legitimate communication.

Need for Firewall

1. Protects systems from unauthorized access.
2. Prevents hackers from entering the network.
3. Blocks malicious traffic and cyber attacks.
4. Controls data entering and leaving the network.
5. Protects sensitive information.
6. Enforces organizational security policies.
7. Monitors network activity.

Objectives of Firewall

1. All traffic between internal and external networks must pass through the firewall.
2. Only authorized traffic should be allowed.
3. The firewall itself must be secure against attacks.

Working of Firewall

When a packet arrives:

1. Firewall examines source address.
2. Checks destination address.
3. Verifies protocol type (TCP, UDP, ICMP).
4. Examines port numbers.
5. Compares information with security rules.
6. Allows or blocks the packet.

Advantages of Firewall

1. Prevents unauthorized access.
2. Protects against hackers.
3. Controls network traffic.
4. Monitors network activities.
5. Enforces security policies.
6. Protects sensitive information.
7. Supports logging and auditing.

Limitations of Firewall

1. Cannot stop attacks originating inside the network.
2. Cannot prevent social engineering attacks.
3. Cannot protect against malware introduced through removable devices.
4. Misconfiguration can create vulnerabilities.
5. Cannot prevent all cyber threats.

Example

Rule:

- Allow HTTP (Port 80)
- Allow HTTPS (Port 443)
- Block Telnet (Port 23)

Result:

- HTTP traffic → Allowed
- HTTPS traffic → Allowed
- Telnet traffic → Blocked

Types of Firewalls

1. **Packet Filtering Firewall:** Packet filtering firewall examines individual packets and decides whether to allow or deny them based on predefined rules.

Parameters Checked

- Source IP Address
- Destination IP Address
- Source Port
- Destination Port
- Protocol Type

Advantages

1. Fast processing.
2. Simple implementation.
3. Low cost.

Disadvantages

1. Cannot inspect packet contents.
2. Vulnerable to IP spoofing.
3. Less secure.

2. **Stateful Inspection Firewall:** A stateful firewall tracks the state of active connections and makes decisions based on the context of traffic.

Working

- Maintains a state table.
- Records active sessions.
- Allows packets belonging to established connections.

Advantages

- More secure than packet filtering.
- Tracks connection states.
- Better attack prevention.

Disadvantages

- Higher memory usage.
- More complex.

3. **Application-Level Gateway (Proxy Firewall):** Acts as an intermediary between client and server.

Working

- Client → Proxy Firewall → Server
- The client never directly communicates with the external server.

Advantages

- High security.
- Content inspection.
- Hides internal network structure.

Disadvantages

- Slower performance.
- Higher processing overhead.

4. **Circuit-Level Gateway:** Operates at the Session Layer and verifies TCP handshakes before establishing connections.

Working

- Validates sessions.
- Does not inspect packet contents.

Advantages

1. Low overhead.
2. Hides internal addresses.

Disadvantages

1. Cannot inspect application data.
2. Limited security.

5. **Next Generation Firewall (NGFW):** Advanced firewall that combines traditional firewall functions with additional security features.

Features

1. Deep Packet Inspection (DPI)
2. Intrusion Prevention System (IPS)
3. Malware Detection
4. Application Awareness
5. URL Filtering

6. SSL Inspection

Advantages

1. Very high security.
2. Real-time threat detection.
3. Protection against modern attacks.

6. **Software Firewall:** Installed on individual computers.

Examples

- Windows Defender Firewall
- Linux iptables

Advantages

1. Easy installation.

2. Individual device protection.

Disadvantages

1. Consumes system resources.
2. Requires maintenance on each device.

7. **Hardware Firewall:** Dedicated physical device placed between network and Internet.

Advantages

1. Protects entire network.
2. High performance.
3. Centralized management.

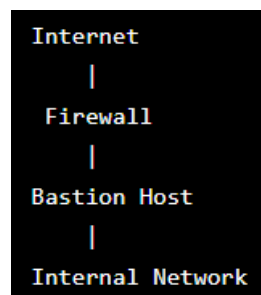
Disadvantages

1. Expensive.
2. Requires configuration expertise

Firewall Configurations

1. **Bastion Host Configuration:** A **Bastion Host** is a highly secured computer that is directly exposed to external networks such as the Internet. It acts as the first point of contact between outside users and the internal network.

- Highly secured operating system.
- Unnecessary services are removed.
- Regular monitoring and updates.
- Strong authentication mechanisms.



Working

1. External users first connect to the Bastion Host.
2. The Bastion Host authenticates and processes requests.
3. Only authorized traffic is allowed to access internal resources.
4. The internal network remains hidden from external users.

Advantages

1. Simple implementation.
2. Provides controlled access.
3. Protects internal systems from direct exposure.

Disadvantages

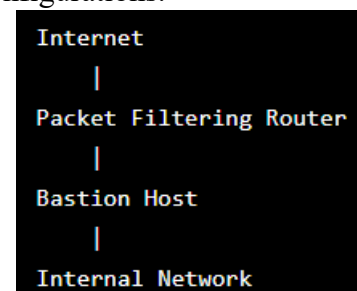
1. Single point of failure.
2. If compromised, security is weakened.
3. Limited protection compared to advanced configurations.

2. **Screened Host Firewall Configuration:** A **Screened Host Firewall System** combines a packet-filtering router (or firewall) with a Bastion Host to provide an additional layer of security.

- Two layers of defense.
- Router filters unwanted traffic.
- Bastion Host provides application-level security.

Working

1. The packet-filtering router examines incoming packets.
2. Only permitted traffic reaches the Bastion Host.
3. The Bastion Host performs additional security checks.
4. Approved traffic is forwarded to the internal network.



Advantages

- Better security than a simple Bastion Host.
- Multiple filtering mechanisms.
- Reduces unauthorized access.

Disadvantages

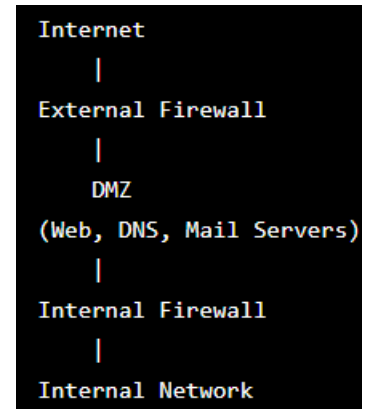
- More complex than a Bastion Host setup.
- Bastion Host remains a critical component.

3. Screened Subnet Firewall Configuration: A Screened Subnet Firewall System uses two firewalls to create a separate network segment known as the **DMZ (Demilitarized Zone)**.

- Uses two firewalls.
- Creates an isolated DMZ.
- Provides layered security.

Working

1. The external firewall filters Internet traffic.
2. Public servers are placed inside the DMZ.
3. The internal firewall protects the private network.
4. Even if a DMZ server is compromised, attackers cannot directly access the internal network.



Advantages

1. Highest level of protection.
2. Public servers are isolated.
3. Internal network remains secure.
4. Suitable for large organizations.

Disadvantages

1. Expensive implementation.
2. Complex configuration and management.

DMZ (Demilitarized Zone) Network

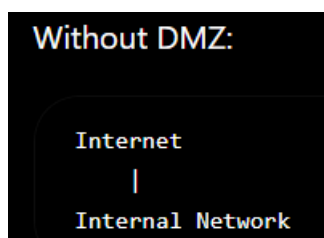
A DMZ (Demilitarized Zone) is a separate network segment positioned between the Internet and the internal network. It contains public-facing servers that must be accessible from the Internet while protecting the internal network.

Systems Usually Placed in DMZ

1. Web Servers
2. Mail Servers
3. DNS Servers
4. FTP Servers
5. Proxy Servers

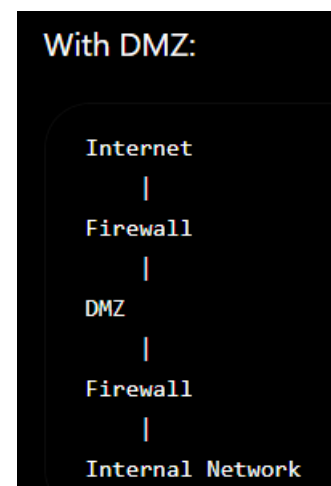
Why DMZ is Needed

1. Provides an additional security layer between the Internet and internal network.
2. Protects sensitive internal systems from direct Internet access.
3. Isolates public-facing servers from the private network.
4. Limits damage if a public server is compromised.
5. Reduces the risk of unauthorized access to internal resources.
6. Allows secure hosting of web, mail, and DNS servers.
7. Enables better monitoring and control of network traffic.
8. Supports implementation of stronger security policies.
9. Prevents attackers from directly reaching internal systems.
10. Improves overall network security and reliability.



Without: If a public server is compromised, attackers can directly access internal systems.

With: Attackers must breach multiple security layers.



Advantages of DMZ

1. Additional security layer.
2. Protects internal network.
3. Limits damage if public server is compromised.
4. Enables secure public services.

5. Improves network segmentation.

Disadvantages of DMZ

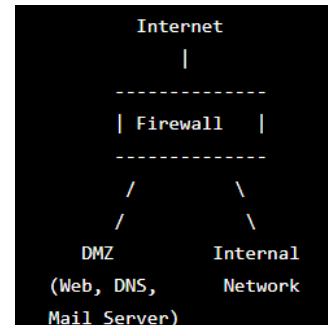
1. Increased complexity.
2. Higher cost.
3. Requires careful management.

DMZ Architecture

1. Single DMZ Architecture: A Single DMZ Architecture uses one firewall with three network interfaces:

- One interface connected to the Internet
- One interface connected to the DMZ
- One interface connected to the Internal Network

The firewall separates the Internet, DMZ, and Internal Network using different security rules.



Working

1. Internet users access public servers located in the DMZ.
2. The firewall filters traffic entering the DMZ.
3. Internal users can access both the Internet and DMZ.
4. Direct Internet access to the internal network is blocked.

Advantages

- Less expensive.
- Easy to configure.
- Requires only one firewall.
- Suitable for small organizations.

Disadvantages

- Single point of failure.
- If the firewall is compromised, the entire network is at risk.
- Lower security compared to Double DMZ.

2. Double DMZ Architecture: A Double DMZ Architecture uses **two firewalls** to create a highly secure DMZ between the Internet and the Internal Network. This configuration is also known as a **Screened Subnet Architecture**.

Working

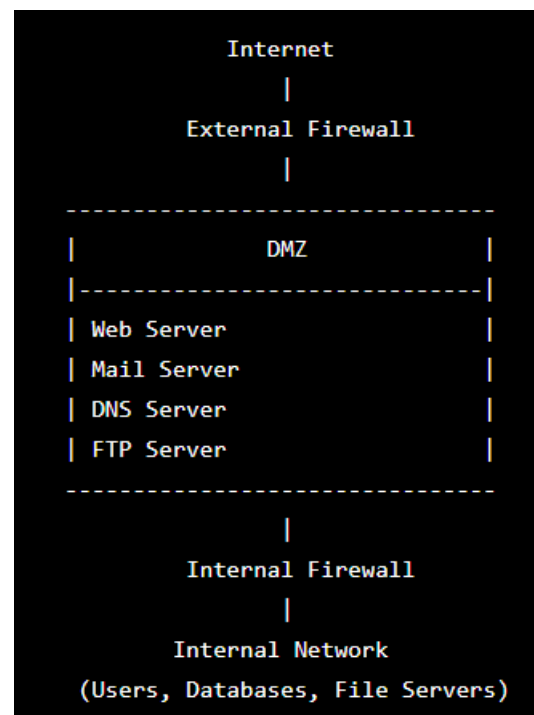
1. Traffic from the Internet first reaches the External Firewall.
2. Allowed traffic enters the DMZ.
3. Public servers provide services from the DMZ.
4. Any traffic moving from the DMZ to the Internal Network must pass through the Internal Firewall.
5. Both firewalls independently inspect and filter traffic.

Advantages

- Very high security.
- Multiple layers of defense.
- Internal network remains protected even if a DMZ server is compromised.
- Suitable for banks, government agencies, and large enterprises.

Disadvantages

- Higher cost.
- More complex management.
- Requires additional hardware and maintenance.



Feature	Single DMZ	Double DMZ
Number of Firewalls	1	2
Security Level	Medium	High
Cost	Low	High
Complexity	Low	High
Maintenance	Easy	Difficult
Single Point of Failure	Yes	No
Suitable For	Small Organizations	Large Organizations
Protection Against Attacks	Moderate	Strong